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Section: 30

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[Assignment-2]

①

a) The seating arrangement is encoded as a permutation of guests 1-8, representing the circular order. Adjacency is defined between consecutive positions, with the first and last guests adjacent.

Four parent chromosomes are:

• Parent 1: [1, 2, 3, 4, 5, 6, 7, 8]

• Parent 2: [1, 3, 5, 7, 2, 4, 6, 8]

• Parent 3: [2, 1, 4, 3, 6, 5, 8, 7]

• Parent 4: [3, 1, 2, 5, 4, 7, 6, 8]

b)

Fitness Function:

For each guest g ,

- +1 for preferred neighbor adjacent to g
- -1 for ~~preferred~~ neighbor adjacent to g avoided

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• preferences and avoidances are defined as:

G1: prefers {2, 3}, Avoids {4}

G2: prefers {1, 4}, Avoids {5}

G3: prefers {1, 6}, Avoids {7}

G4: prefers {2, 5}, Avoids {8}

G5: prefers {4, 6}, Avoids {1}

G6: prefers {3, 5}, Avoids {2}

G7: prefers {8}, Avoids {3}

G8: prefers {7}, Avoids {4}

Fitness Calculation:

• Parent 1:

G1: +1 (adjacent to 2)

G2: +1 (adjacent to 1)

G3: 0

G4: +1 (adjacent to 5)

G5: +2 (adjacent to 4 & 6)

G6: +1 (adjacent to 5)

G7: +1 (adjacent to 8)

G8: +1 (adjacent to 7)

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$$\therefore \text{Fitness} = 1 + 1 + 0 + 1 + 2 + 1 + 1 + 1 = 8$$

Similarly,

• parent 2 : $\text{Fitness} = 1 + 1 + 1 + 1 = 4$

• parent 3 : $\text{Fitness} = 1 + 1 + (-1) + 1 + 1 + 1 + 1 + 1 + 1 = 7$

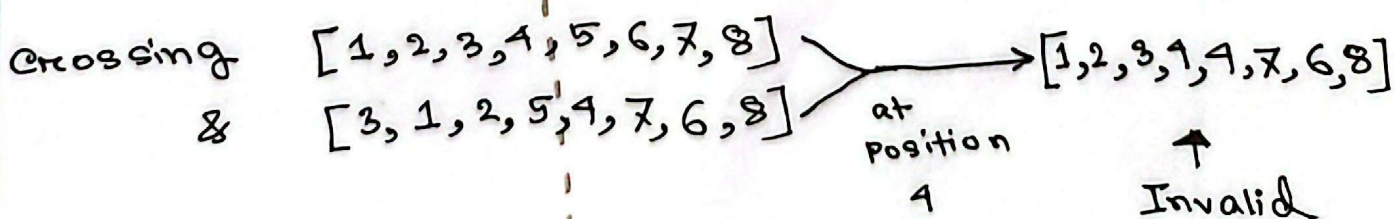
• parent 4 : $\text{Fitness} = 1 + 1 + 1 + 1 + (-1) + 1 + 1 = 5$

Two fittest chromosomes :

• parent 1 (Fitness = 8)

• parent 3 (Fitness = 7)

c) Regular crossover (one-point cross over) will not work because it produces duplicate/missing guests. For example,



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In the n-queens problem : GA can get away with it because genes are independent, but here each gene must be a unique guest. So, we need a permutation preserving operator.

2) If we swap two randomly selected positions in a chromosome, that might be a possible case of mutation - in this context.

Ex: Parent 4 = [3, 1, 2, 5, 4, 7, 6, 8]

After

Mutation

[3, 1, 4, 5, 2, 7, 6, 8]

GOOD LUCK

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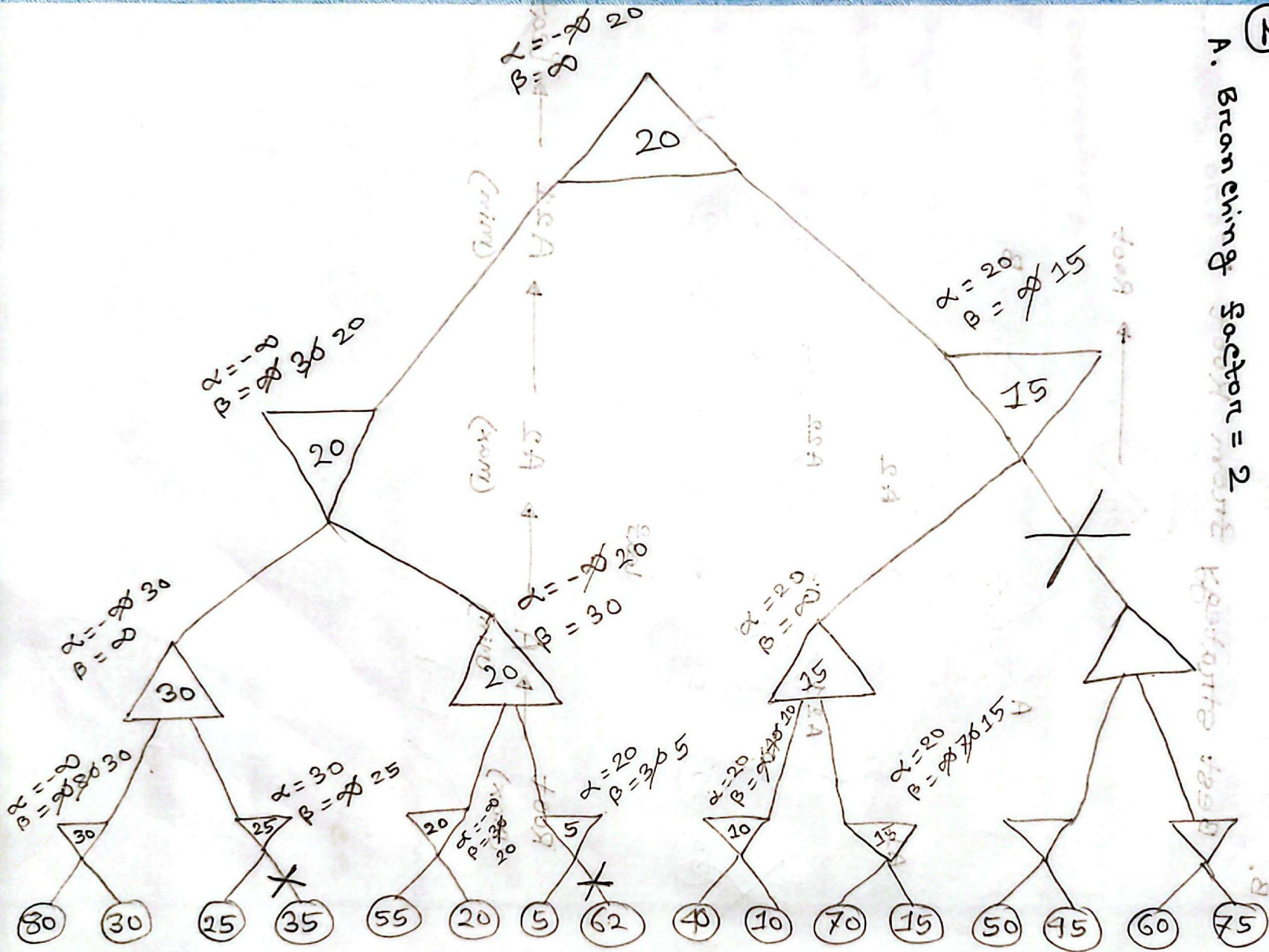
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A. Branching factor = 2



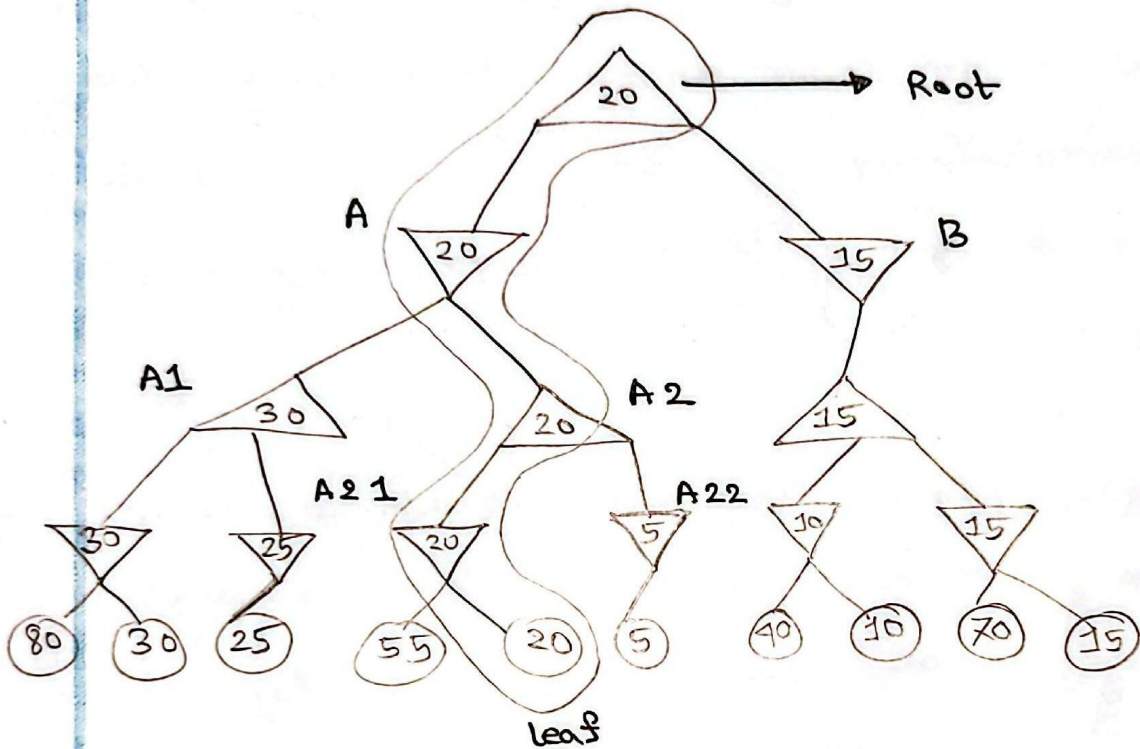
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B.

Best strategy from Root :-



Root (max) $\rightarrow$ A (min) $\rightarrow$ A2 (max) $\rightarrow$ A21 (min) $\rightarrow$ leaf (20)

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Q.

Yes, ordering improves performance.

Descending order : At max nodes.

Evaluate children from highest to lowest value. A high value found early raises α . We can get the max value. Subsequent children are pruned if their values cannot exceed α .

Ex -

max node children values [50, 80, 20]

Descending order : [80, 50, 20]

Evaluate 80 first $\rightarrow \alpha$ becomes 80

Next ^{child} 50 is pruned (since $80 > 50$)

child 20 is never evaluated.

So, 1 node evaluated instead of 3

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Ascending order: At min nodes.

Evaluate children from lowest to highest value.

A lower value found early lowers β .
we can get the best min value.
subsequent children are pruned if their value cannot be lower than β .

Ex -

Min node children values $[40, 10, 70]$

Ascending order $[10, 40, 70]$

Evaluate 10 first $\rightarrow \beta$ becomes 10

Next child 40 is pruned $\rightarrow$ (since $40 > 10$)

child 70 is never evaluated.

So, 1 node evaluated instead of 3.